

Homework 7

2.3.5 Let $g = t^{-1}\chi_{t \geq 1}$ defined on $\mathbb{R}$. Prove that for $|\xi| \leq 1$ we have

$$|\widehat{g}(\xi)| \leq \frac{1}{\pi} + \ln\left(\frac{1}{|\xi|}\right),$$

while $|\widehat{g}(\xi)| \leq (\pi|\xi|)^{-1}$ for $|\xi| \geq 1$.

Proof. Clearly, $g(t) = t^{-1}\chi_{t \geq 1} \in L^2(\mathbb{R}) \setminus L^1(\mathbb{R})$. Consider the sequence $g_N(t) = g(t)\chi_{t \leq N}$. Then each $g_N(t)$ is in $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, and $g_N \rightarrow g$ in $L^2(\mathbb{R})$. Thus, $\widehat{g_N} \rightarrow \widehat{g}$ in $L^2(\mathbb{R})$. We must show that for all $\xi \neq 0$,

$$\lim_{N \rightarrow \infty} \int_1^N g(t) e^{-2\pi i t \xi} dt = \int_1^\infty g(t) e^{-2\pi i t \xi} dt \quad (1)$$

exists. According to proposition 2.3.6, $\widehat{g} = \widetilde{\widehat{g}}$. However, because g is real valued, we get that $\widehat{g}(\xi) = \widetilde{\widehat{g}}(-\xi)$. If $\xi < 0$, then $\widehat{g}(\xi) = \widetilde{\widehat{g}}(|\xi|)$, and so $|\widehat{g}(\xi)| = |\widetilde{\widehat{g}}(|\xi|)| = |\widehat{g}(|\xi|)|$. Thus, we can replace ξ with $|\xi|$. We now do a change of variables on (1). Let $s = t\xi$. This gives $t = s/\xi$, $dt = ds/\xi$. show

$$\widehat{g}(|\xi|) = \int_{|\xi|}^\infty \frac{e^{-2\pi i s}}{s} \cdot \frac{\xi}{\xi} ds = \int_{|\xi|}^\infty \frac{e^{-2\pi i s}}{s} ds. \quad (2)$$

Integration by parts yields:

$$\begin{aligned} \widehat{g}(|\xi|) &= \lim_{\rho \rightarrow \infty} \left[\frac{-e^{-2\pi i s}}{2\pi i s} \right]_{|\xi|}^\rho - \int_{|\xi|}^\infty \frac{e^{-2\pi i s}}{2\pi i s^2} ds \\ &= \frac{e^{-2\pi i |\xi|}}{2\pi i |\xi|} - \int_{|\xi|}^\infty \frac{e^{-2\pi i s}}{2\pi i s^2} ds \\ &= \frac{1}{2\pi i} \left[\frac{e^{-2\pi i |\xi|}}{|\xi|} - \int_{|\xi|}^\infty \frac{e^{-2\pi i s}}{s^2} ds \right] \\ \Rightarrow |\widehat{g}(\xi)| &\leq \frac{1}{2\pi} \left[\left| \frac{e^{-2\pi i |\xi|}}{|\xi|} \right| + \left| \int_{|\xi|}^\infty \frac{e^{-2\pi i s}}{s^2} ds \right| \right] \\ &\leq \frac{1}{2\pi} \left[\frac{1}{|\xi|} + \int_{|\xi|}^\infty \left| \frac{e^{-2\pi i s}}{s^2} \right| ds \right] \\ &= \frac{1}{2\pi} \left[\frac{2}{|\xi|} \right] = \frac{1}{\pi|\xi|}. \end{aligned}$$

When $|\xi| \geq 1$. Now, let $|\xi| \leq 1$. We split (2) at 1. Notice that the second integral is what we bounded above when $|\xi| = 1$.

$$\widehat{g}(|\xi|) = \int_{|\xi|}^1 \frac{e^{-2\pi i s}}{s} ds + \int_1^\infty \frac{e^{-2\pi i s}}{s} ds \leq \int_{|\xi|}^1 \frac{e^{-2\pi i s}}{s} ds + \frac{1}{\pi}.$$

The absolute value of the remaining integral is bounded by the integral of $1/s$, which evaluates to $\ln(1) - \ln(|\xi|) = 0 + \ln(\frac{1}{|\xi|})$. So, $|\widehat{g}(\xi)| \leq \frac{1}{\pi} + \ln(\frac{1}{|\xi|})$ as desired. □